

R4: Uncertainty in the Known Draught γ

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How accurately must the floe draught be known?

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SHARC buoy project Code: `runR4GammaTolerance.m`, commit 1147d1f + R4 config Results: `Results/R4/R4_20261001_183822`

Summary

Question. The frozen inverse estimates $[\ln \beta, \ln R]$ and fixes the draught parameter γ at an assumed value. If that value is wrong by ϵ_γ , the fit absorbs the error through β and R . How large can ϵ_γ be? **Outcome.** A wrong γ leaks almost entirely into stiffness: per unit error in $\ln \gamma$, $\ln \hat{\beta}$ moves by 0.090 and $\ln \hat{R}$ by only 0.008 (LSM6DSV16X). Keeping the stiffness error below 1% needs $|\epsilon_\gamma| \leq 10.6\%$ (± 0.29 mm of draught at p_0). Keeping the bias below half a Fisher uncertainty unit ($d_{\text{sys}} \leq 0.5$) needs $|\epsilon_\gamma| \leq 1.64\%$ (± 0.045 mm), and this requirement tightens as the sensor gets quieter. Exact-EMM checks confirm the linear projection to 0.03σ at $\pm 5\%$. **Practical consequence.** ± 0.045 mm cannot be measured with a ruler, but γ is mass per unit area, so it should be obtained by *weighing* the floe, which easily reaches 1%.

1 Question and setting

The draught parameter is $\gamma = d/H = \rho_i h / (\rho_w H)$. The inverse uses an assumed value

$$\gamma_{\text{inv}} = \gamma_{\text{true}}(1 + \epsilon_\gamma),$$

so a relative error in γ is the same relative error in the draught d (or in the thickness h , with ρ_i known). The study is evaluated at p_0 with the E1 production setting: Level 2C layout with the reference at $0.9R$, $N = 65536$ at $\Delta t = 0.1$ s, Hann $L = 2048$, 84 bins, JONSWAP $H_s = 0.05$ m, $\omega_p = 5$ rad/s. All six pre-registered noise cases are evaluated; the headline case is the LSM6DSV16X. At p_0 the draught is $\gamma_0 H = 2.74$ mm.

2 Method

1. Sensitivity to γ . E1 computes $J_\gamma = \partial f / \partial \ln \gamma$ by central differences (two extra EMM fields) as a *nuisance* sensitivity. It never enters the Fisher matrix F , which stays 2×2 .

2. Linear projection. Data from the true floe compared with a model using the wrong γ gives the feature bias

$$b_\gamma = f(\gamma_{\text{true}}) - f(\gamma_{\text{inv}}) \approx -J_\gamma \ln(1 + \epsilon_\gamma),$$

which the E1 bias projector maps to a parameter shift and its significance:

$$\delta \theta_\gamma = F^{-1} J^T \Sigma^{-1} b_\gamma, \quad d_{\text{sys}} = \sqrt{\delta \theta_\gamma^T F \delta \theta_\gamma}.$$

Each noise case uses its own Σ_k , F and accepted-bin mask. The pre-registered grid is $\epsilon_\gamma = \pm 0.5, \pm 1, \pm 2, \pm 5, \pm 10, \pm 20\%$.

3. Tolerances. On a fine grid ($\pm 30\%$, 1201 points) the smallest $|\epsilon_\gamma|$, over both signs, at which each criterion is first reached:

Criterion	Meaning	Type
$ \delta \beta / \beta = 1\%$ (with $\delta \beta / \beta = e^{\delta \ln \beta} - 1$)	stiffness has moved by 1% physically	accuracy
$d_{\text{sys}} = 0.5$	bias is half the random-uncertainty scale	significance
$d_{\text{sys}} = 1$	bias is as large as the random uncertainty	significance

4. Nonlinear model check (headline noise, $\epsilon_\gamma = \pm 5\%$ and $\pm 20\%$). The EMM is solved exactly at γ_{inv} and the wrong- γ model is fitted to the noiseless true features with a fixed-Jacobian (chord) Gauss-Newton iteration until the step is below 0.01σ . This tests whether the linear J_γ projection holds at the errors that matter. It is a model check, not an independent noisy inversion.

3 Results

Tolerances by noise case.

Noise case	$\sigma_{\ln \beta}$	d_W	per unit $\ln \gamma$ error		tolerance $ \epsilon_\gamma $			$d_{\text{sys}} \leq 0.5$ draught (mm)
			$\delta \ln \beta$	$\delta \ln R$	$ \delta \beta / \beta \leq 1\%$	$d_{\text{sys}} \leq 0.5$	$d_{\text{sys}} \leq 1$	
Datasheet grade	0.12%	0.624	0.0834	0.00694	11.35%	0.60%	1.19%	± 0.016
Three IMUs fused	0.28%	0.369	0.0888	0.00799	10.70%	1.33%	2.65%	± 0.036
LSM6DSV16X	0.36%	0.309	0.0896	0.00813	10.61%	1.64%	3.26%	± 0.045
G3 assumption	0.47%	0.244	0.0902	0.00826	10.54%	2.13%	4.22%	± 0.058
ICM-42688-P	0.48%	0.236	0.0903	0.00827	10.53%	2.21%	4.36%	± 0.060
LSM6DS3TR-C	1.38%	0.088	0.0913	0.00842	10.43%	6.04%	11.71%	± 0.165

Rows ordered from quietest to noisiest. The 1% stiffness tolerance corresponds to ± 0.29 mm of draught for every case (0.285 to 0.310 mm). Draught values are at p_0 (2.74 mm).

Nonlinear model check (LSM6DSV16X).

ϵ_γ	$\delta\beta/\beta$		nonlinear / linear -1		difference [$\ln \beta, \ln R$] in σ	exact vs linear feature error
	linear	nonlinear	$\ln \beta$	$\ln R$		
-20%	-1.979%	-1.777%	-10.3%	-10.9%	[+0.58, +0.29]	10.5%
-5%	-0.458%	-0.447%	-2.5%	-2.7%	[+0.03, +0.02]	2.6%
+5%	+0.438%	+0.449%	+2.5%	+2.6%	[+0.03, +0.02]	2.5%
+20%	+1.647%	+1.806%	+9.6%	+10.3%	[+0.44, +0.23]	9.9%

Each case converged in 2 to 3 iterations. The relative error grows roughly linearly with $|\epsilon_\gamma|$ (about 0.5% per 1% of γ error), and in both directions the exact shift lies closer to the positive side.

Figure: Results/R4/figures/R4_fig1.gamma_tolerance.pdf (copy it next to this file).

Figure 1. (a) Induced stiffness and radius errors against the error in the assumed γ (LSM6DSV16X), linear projection with the exact-EMM check (open circles); dashed lines at $\pm 1\%$. (b) Bias significance d_{sys} for every noise case; dashed lines at $d = 0.5$ and $d = 1$.

4 Interpretation

1. **A wrong draught biases stiffness, not radius.** $\delta \ln \beta / \delta \ln R \approx 11$. A 10% error in γ moves β by about 0.9% and R by about 0.08%. Physically, γ sets the plate's inertia, which the fit compensates mainly by changing its stiffness.
2. **Two criteria answer two different questions.** The 1% criterion asks whether β has physically moved by 1%; it depends only on the model geometry, so it is almost the same for every noise case (10.4 to 11.4%). The d_{sys} criterion asks whether the shift is large compared with how precisely β can be estimated; it varies tenfold (0.60 to 6.04%). The noisy LSM6DS3TR-C does *not* tolerate draught error better: its random uncertainty is simply large enough to hide the bias.
3. **The tolerances are trustworthy where they matter.** The d_{sys} tolerances for the realistic sensors (1.3 to 6%) lie inside $\pm 5\%$, where the exact EMM agrees with the linear projection to 0.03σ . The 1% stiffness tolerance (10.6%) lies between the checks. It is reached first on the negative side, where the exact shift is about 5% *smaller* than the linear one at $\epsilon_\gamma \approx -10\%$, so the linear 10.6% is slightly conservative (the exact value is about 11%).
4. **Better sensors expose estimator bias.** With γ known exactly, the Welch-resolution bias alone gives $d_W = 0.624$ for the datasheet-grade case, above the Identifiable limit of 0.5: at $L = 2048$ that case would already be Marginal. As the sensor improves, the limiting error moves from random noise, to draught uncertainty, to the estimator's own resolution bias. For the realistic sensors d_W is 0.09 to 0.37.
5. **How to meet the requirement in practice.** ± 0.045 mm of draught cannot be measured directly. But

$$\gamma = \frac{\rho_i h}{\rho_w H} = \frac{m}{\rho_w \pi R_{\text{phys}}^2 H}, \quad \left(\frac{\delta \gamma}{\gamma}\right)^2 \approx \left(\frac{\delta m}{m}\right)^2 + \left(2 \frac{\delta R_{\text{phys}}}{R_{\text{phys}}}\right)^2 + \left(\frac{\delta \rho_w}{\rho_w}\right)^2.$$

Weighing to 0.5% and measuring the radius to ± 2 mm (0.3%) gives about 0.8%, inside the LSM6DSV16X $d_{\text{sys}} \leq 0.5$ tolerance. γ should therefore be determined from the floe's mass and area, not from a measured draught.

6. **Instrument mass is a likely limiting term (open item).** The inverse assumes uniform areal mass. The basin plate at p_0 is about 4.4 kg. If each sensor package were about 50 g (not yet weighed), four would add about 4.5% of the plate mass, beyond the 1.64% tolerance if treated as a uniform γ error, and in reality concentrated at four points rather than spread. This must be quantified once the packages are weighed: either include the instrument mass in γ , or model it as point masses.

5 Limitations

- **One floe.** Evaluated at p_0 only. Other floes have other sensitivities; A1 reports them per floe.
- **d_{sys} alone.** The tolerances use the draught bias alone. Classification uses $d_{\text{tot}} = \|\delta\theta_W + \delta\theta_{\text{sys}}\|_F$, where the two biases add as vectors, so the combined tolerance depends on their relative direction and differs between signs of ϵ_γ . The per- ϵ classes with d_{tot} are in R4_sweep.csv.
- **Fixed layout and sea state.** Level 2C and the U1 JONSWAP spectrum only.
- **Comparison with the earlier estimate.** An earlier single-FRF magnitude estimate gave 0.057 for β and 0.008 for R . The radius coupling agrees; the stiffness coupling is 1.6 times larger in the full transmissibility and Fisher treatment. The two calculations use different observables, so this is a consistency check, not a validation.

Requirement for the experiment and the thesis

For the LSM6DSV16X at p_0 : γ to within $\pm 1.6\%$ keeps the draught-induced bias below half an uncertainty unit; within $\pm 10.6\%$ keeps the stiffness error below 1%. Determine γ by weighing (mass and area), include the instrument mass, and report γ with its uncertainty alongside every recovered β .

Output	Contents
R4_20261001.183822.log / .mat / _summary.txt	full command-window log; results and config; headline numbers
R4_tolerance.csv	one row per noise case: σ , sensitivities, three tolerances
R4_sweep.csv	72 rows (6 noise cases $\times$ 12 ϵ_γ): $\delta\beta/\beta$, $\delta R/R$, d_{sys} , d_{tot} , class
figures/R4_fig1_gamma_tolerance.pdf/.png	Figure 1 (layout to be tidied before the thesis)

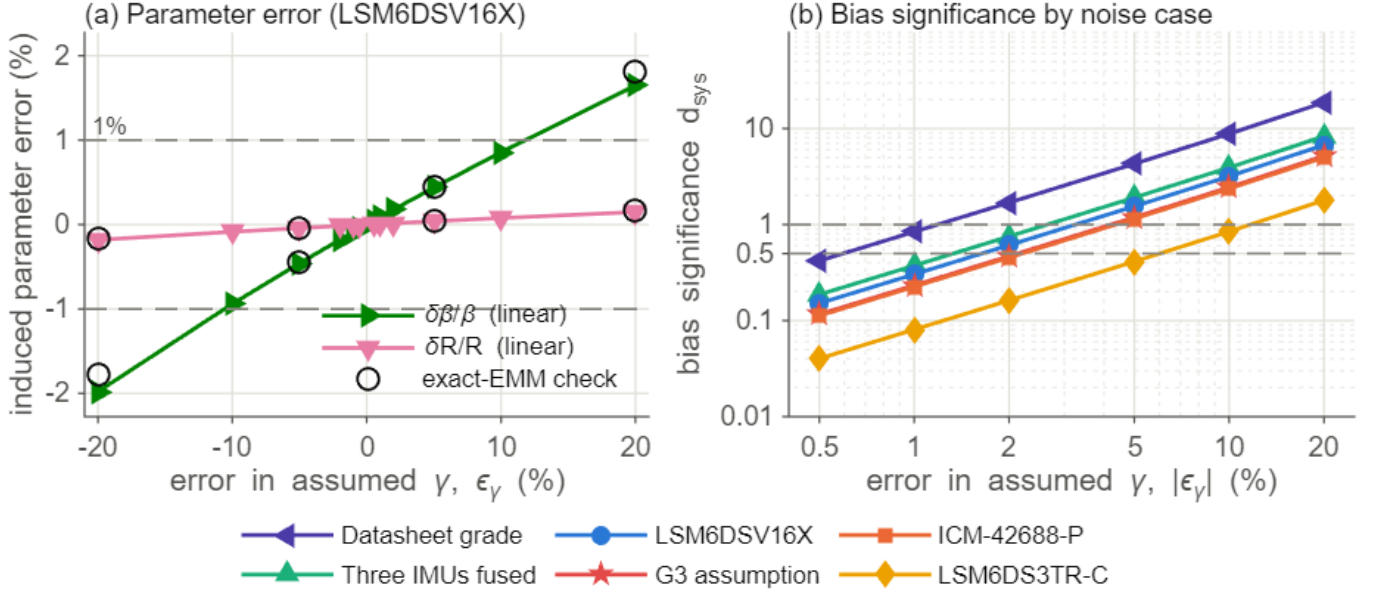


Figure 1: R4 sensitivity of the recovered parameters to uncertainty in the known draught parameter γ . (a) Induced relative errors in β and R for the LSM6DSV16X noise case, with exact-EMM nonlinear check points shown as open circles and $\pm 1\%$ stiffness-error thresholds indicated by dashed lines. (b) Systematic bias significance d_{sys} versus the magnitude of the assumed γ error for all noise cases; dashed lines indicate the $d_{\text{sys}} = 0.5$ and $d_{\text{sys}} = 1$ tolerance levels.